

Taneem Ullah Jan

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Applied AI researcher and engineer building production-grade AI systems across LLMs, agentic workflows, and multimodal learning: from research to reliable real-world deployment.

PROFESSIONAL EXPERIENCE

AI Lead

Oct. 2025 – Present

ErisAI

- Lead the research, architecture, and engineering of production-grade AI systems spanning LLMs, multimodal models, retrieval, memory, and agentic workflows.
- Architect and develop **Synapse**, a research-driven AI infrastructure layer for multi-agent workflows, integrating hybrid retrieval, multi-turn memory, intelligent routing, and model orchestration for reliable real-world deployments.
- Define evaluation protocols and operational controls for response quality, traceability, knowledge access, and failure analysis across enterprise AI systems.
- Deployed **Synapse** across multiple enterprise AI applications, including legal and investigative platforms, processing over 30M+ private documents with grounded retrieval and traceable responses. ([Use case](#))

AI Researcher

Sep. 2024 – Oct. 2025

VOLV AI

- Researched and developed a 3D virtual try-on SDK using neural rendering, GANs, and diffusion models; the system was adopted by fashion brands and increased user engagement by 40%. ([Demo](#))
- Developed controllable 2D and 3D digital human systems with pose estimation and generative modeling, supporting personalized experiences and reducing product return rates by 30%.
- Prototyped multimodal LLM systems integrating language, vision, and audio for cross-modal reasoning and instruction-following in customer support and interactive applications.

Research AI Engineer

Jan. 2023 – Dec. 2023

BHuman AI

- Built scalable generative video pipelines using neural reenactment, facial motion transfer, and voice cloning, reducing production costs by 40% for a platform serving 90K+ users and enterprise media clients. ([Use cases](#))
- Researched and deployed audio-driven lip-sync models using GANs and super-resolution, improving visual fidelity and establishing a core product capability and primary revenue stream.
- Integrated LLMs with persona-driven avatars and developed retrieval-based conversational systems for enterprise clients, combining private knowledge, persona conditioning, and response evaluation for real-world deployments.

Intern Machine Learning Engineer

Aug. 2021 – Nov. 2021

NAECO Blue GmbH

[[Profile](#)]

- Developed machine learning models to improve the spatial and temporal resolution of weather data for solar and wind energy forecasting.
- Built a data pipeline that reduced R&D time by nearly 50% and supported forecasting workflows later adopted by 5+ European meteorological agencies.

PUBLICATIONS

Taneem, U. J., & Ayesha, N. (2024). [Beyond CNNs: Encoded Context for Image Inpainting with LSTMs and Pixel CNNs](#). International Journal of Innovations in Science & Technology, (IJIST) 6(5), 165–179, Special Issue ICTIS 2024.

Taneem, U. J., & Zakira, I. (2022). [HTML Code Generation from Images with Deep Neural Networks](#). Journal of Engineering and Applied Sciences (JEAS), UET Peshawar. (**Young Undergraduate Researcher Award**)

SELECTED PROJECTS

Synapse (2026). [Research-driven AI system integrating retrieval, memory, routing, evaluation, and runtime controls for reliable model behavior.](#)

Ownify AI (2026). [Multi-tenant AI platform integrating knowledge isolation, retrieval, conversational memory, orchestration, and operational infrastructure for dependable production AI systems.](#)

PatchProof (2026). [Framework for evidence-grounded patch verification using execution traces, behavioral validation, and correctness analysis beyond unit tests.](#)

DarwinPatch (2026). [Verifier-grounded coding-agent repair framework combining bounded failure evidence, route-aware search, regression gates, and auditable benchmark evaluation.](#)

DGM-LLM (2025). [Autonomous code self-improvement framework combining evolutionary search with LLM-guided mutation and multi-dimensional evaluation.](#)

VMVLM (2025). [Vision-language architecture using direct visual feature modulation to improve multimodal reasoning and instruction-following.](#)

EmbedVoiceLLM (2025). [Efficient multimodal framework mapping speech representations directly into LLM embedding space for low-latency voice interaction.](#)

FlexiSMPL (2025). [Real-time SMPL body modeling system with measurement-driven control and interactive 3D visualization.](#)

SKILLS

- **Programming:** Python, C++, Bash, MATLAB
- **ML / AI:** PyTorch, Transformers, TensorFlow, NumPy, multimodal learning, fine-tuning, model evaluation, experiment design
- **LLM Systems:** RAG, Hybrid Retrieval, Vector Search, Conversation Memory, Agentic Workflows, LLM Orchestration, vLLM
- **Frameworks:** LangChain, Haystack, OpenAI API, Anthropic API, Gemini API
- **Vector Databases:** Qdrant, FAISS
- **Computer Vision / 3D:** OpenCV, PyTorch3D, MediaPipe, ONNX
- **Infrastructure:** Docker, Git, Linux, Weights & Biases, AWS, Azure, GCP

EDUCATION

University of Engineering and Technology Peshawar, Pakistan

Bachelor of Science in Computer Science, CGPA: 3.58/4.0

Sep. 2018 – Sep. 2022